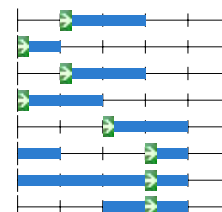


Program overview

19-May-2025 10:51

Year 2024/2025
Organization Civil Engineering and Geosciences
Education EngD Civil and Environmental Engineering

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
EngD CEE 2024		Short EngD Civil and Environmental Engineering 2024					
PDEng CEE Obligatory courses							
CEE5000	Effective Communication for Designers	2					
CEE5001	Research Design	2					
CEE5002	Integrated Design	4					
CEE5003	Asset Management for Designers	4					
CEE5004	Risk Management for Designers	4					
CEE5005	Group Design Project	12					
CEE5006	Individual Design Project	60					
CEE5007	Techno-Economic Evaluation for Designers	6					
PDEng CEE Elective courses							



Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	EngD Civil and Environmental Engineering

EngD CEE 2024

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	EngD Civil and Environmental Engineering

PDEng CEE Obligatory courses

CEE5000	Effective Communication for Designers	2
Responsible Instructor	Ing. C.H. Groenewegen	
Instructor	S. Cornelissen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	2 3	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	This course helps designers and engineers to learn the basics of effective communication with different types of audiences and to present ideas, models or new technology to management and senior staff. Studies have shown that the audience digresses from the topic in technical presentations and speeches that take longer than 6-7 minutes. It seems that being concise and having a clear message is very difficult for most presenters with an engineering background. Engineers and scientists prefer to show how much they have done rather than how the obtained results relate to main topic, for instance by showing a wide range of results and graphics. This causes unnecessary long presentations with an audience that get lost in details, get confused and stray from the main topic. In this course, the attendees will also learn to structure the information in a way the target audience will understand the main message of the speaker easily. In addition, attendees learn to be efficient in time and focusing on the main topic rather than details and also learn to get in contact with audience and interact efficiently to deliver the main message.	
Study Goals	* Identify the characteristics of the target group for a message or presentation * Adapt the structure and content of a presentation or message to the level and concerns of the audience * Compose a clear story line using suitable metaphors and examples , and a conclusion that fits with the objective of the presentation * Prepare presentations with a size, duration and level of complexity appropriate to the situation. * Verify the effect of the message at the audience during the presentation and adapt the performance instantly * Use techniques and methods to convey a message effectively to the audience * Use techniques for effective responses to questions and opposing opinions * Support a case and bring about arguments effectively in meetings and discussions	
Education Method	* 5 compulsory workshop days, mixed between live/online. * Assignments and self-study before every workshop (~2-3 h per workshop) * Building and giving a final presentation for a group of experts.	
Assessment	* Attendance and performance during lectures (20% of the grade) * Preparing and giving a long presentation (15-20 minutes) on the individual project after the course (80% of the grade) to a team of experts	
Remarks	In principle this course is for participants of the EngD CEE programme only. If you are a PhD student and would like to follow this course please ask the responsible instructor if you can join this course.	
Expected prior Knowledge	MSc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	Handouts	
Judgement	see Assessment	
Permitted Materials during Exam	laptop, beamer	
Collegerama	No	

CEE5001	Research Design	2
Responsible Instructor	Dr.ir. M. van den Boomen	
Course Coordinator	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	1	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Design a scientifically sound research approach for the engineering objective of your individual project and write this down in your design brief	
Study Goals	After the course you should be able to: 1. Describe the context of the engineering objective and its relevance; 2. Describe the present state of the relevant technology and scientific knowledge; 3. Identify the research gap or contribution that is addressed by the engineering objective; 4. Define the research objective, research questions and expected result; 5. Select appropriate research methods and their validation to answer the research questions and obtain the expected results; 6. Improve the research design of the EngD documents in writing.	
Education Method	Short introductions to the themes and the projects of the participants; Plenary and group presentations of the research question or design problem, key variables or design brief, hypothesis or prototype/working principle, method; Providing and processing (peer)feedback.	
Assessment	The trainees research design for the EngD definition phase (report) is assessed.	
Remarks	In principle this course is for participants of the EngD CEE only. If you are a PhD student and would like to follow this course please ask the responsible instructor if you can join this course.	
Expected prior Knowledge	MSc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	Course reader, handouts, lecture slides and selected references for literature.	
Judgement	See Assessment	
Permitted Materials during Exam	None	
Collegerama	No	

CEE5002	Integrated Design	4
Responsible Instructor	Ir. W.G. van der Hoog	
Course Coordinator	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	2 3	
Start Education	2	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	Integrated Design (CEE 5002) is a compulsory course of the EngD CEE programme. During this course, the trainee needs to develop design innovation skills and stakeholder involvement skills. In addition to the learning objectives, the course includes learning objectives related to personal development such as design skills and effective communication. The result needs to match with the requirements of the involved company and individual project. The trainees will go through a design innovation training trajectory spread over 6 half days and will apply the knowledge and skills directly in their individual projects. In the program, each trainee works on his/her individual project with the aim to realize a working prototype. In this course, the trainees are supported in creating a better design. They will learn how to go from an initial idea to a solution with impact. The design needs to be desirable, viable and feasible. The written work and presentation	
Study Goals	Learning objectives related to integrated design knowledge development and application: · To acquire knowledge of models, methods, and techniques for analyzing stakeholders; · To determine role and effect of stakeholders on the design process; · To develop requirements and criteria for the design of a system, component or method considering several stakeholders; · To determine the effect of proposed design on internal and external stakeholders. Learning objectives related to the trainees individual project: · To identify project requirements, their types and their importance in the trainees individual project; · To draft an appropriate stakeholder/social relevance research question and research design which contributes to the trainees individual project; · To involve relevant stakeholders through interviews and creative sessions; · To analyze and integrate adjusted and or new requirements for further improvements of the design. Learning objectives related to skills: · Visualizations · Prototyping · Effective communication · Involving stakeholders · Design Thinking	
Education Method	Lectures, workshop and assignments	
Assessment	A report on the creation of a prototype of new product or a solution to a problem in the organisation of the participant. The participant will also present the prototype and the various steps in the creation process. Report and presentation will be judged on understanding of the theory and the usability/feasibility of the prototype.	
Remarks	In principle this course is for participants of the EngD CEE only. If you are a PhD student and would like to follow this course please ask the responsible instructor if you can join this course.	
Expected prior Knowledge	MSc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	Handouts and book	
Judgement	See Assessment	
Permitted Materials during Exam	None	
Collegerama	No	

CEE5003	Asset Management for Designers	4
Responsible Instructor	Dr.ir. M. van den Boomen	
Course Coordinator	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	1 2	
Start Education	1	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The purpose of the course CEE5003 is to learn to integrate asset management tools and methods for optimizing design, operation, maintenance and replacements of physical assets (systems and components) within the individual applied research project. In this course fundamental tools and methods required for this optimization are lectured such as reliability engineering, reliability centered maintenance, monitoring and data handling, and life cycle cost analysis.	
Study Goals	After this course you should be able to integrate asset management tooling and methods to optimize the design and life cycle management of equipment and infrastructures. More specifically you should be able to: 1. Design an integrated asset management strategy that contributes to your individual engineering objective 2. Execute an integrated asset management strategy that contributes to your individual engineering objective 3. Demonstrate designer skills that correspond with the EngD programme requirements (applied research)	
Education Method	Lectures, individual counseling, individual assignment	
Assessment	A research paper or report in which the developed asset management strategy or method is executed in the trainees individual project and organization. This will be defended in a 15-minute presentation on the essentials of the research. The paper and presentation will be judged on the understanding of the theory, the research methodology and its execution, and the ability to make recommendations for improvements within the trainees individual project and organization on the identified asset management challenges	
Remarks	In principle this course is for participants of the EngD CEE only. If you are a PhD student and would like to follow this course please ask the responsible instructor if you can join this course.	
Expected prior Knowledge	Msc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	Course reader and handouts	
Judgement	See Assessment	
Permitted Materials during Exam	None	
Collegerama	No	

CEE5004	Risk Management for Designers	4
Responsible Instructor	Prof.dr.ir. P.H.A.J.M. van Gelder	
Course Coordinator	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	3 4	
Start Education	3	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The future cannot be predicted. It is uncertain and no one has ever been successful in consistently forecasting the stock market, interest rates, exchange rates, or commodity prices or credit, operational, and systemic events with major financial implications. Risk management is considered as a tool to monitor, minimize and control the impact of unfortunate events through the identification, evaluation, and prioritization of risks followed by coordinated and economical application of resources. * General introduction to risk management * The steps in a risk management process * Types of risks with techniques and methods for risk detection * Approaches and instruments for risk quantification * Models and types of analysis to aggregate risks * Financial, operational and strategic techniques to control, risk; * Methods and instruments for investigation material related damages * Damages in various materials	
Study Goals	* An understanding of the sequential steps of a risk management process (Establishing context, Identification, Analyzing, Integration, Assessing, Treating and Monitoring) in the environment of the individual design project. * Understand and apply appropriate risk quantification approaches * Identify and quantify hazard risks, financial risks, operational risks and strategic risks for the organization involved in the individual design project. * Aggregating and assessing all risk distributions, reflecting correlations and effects, and expressing the impact on key performance indicators that are related to the individual project. * Developing approaches and measures for controlling and monitoring of the identified risks * Understand the relevant structural failure mechanisms in various materials * Develop measures to deal with the failure mechanisms * Understand the legal aspects that are relevant for liability after occurrence of damage	
Education Method	Lecturers, workshop and assignments	
Assessment	A report about a risk assessment in an environment or organization related to the individual project. The participant will present the steps of the risk management process and come with measures for controlling the identified risks. Report and presentation will be judged on understanding of the theory and the appropriate measures.	
Remarks	In principle this course is for participants of the EngD CEE only. If you are a PhD student and would like to follow this course please ask the responsible instructor if you can join this course.	
Expected prior Knowledge	MSc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	Handouts	
Judgement	See Assessment	
Permitted Materials during Exam	None	
Collegerama	No	

CEE5005	Group Design Project	12
Responsible Instructor	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	1 4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The group design project is a collective assignment of the EngD programme Civil and Environmental Engineering on a real-life design question in the civil industry. The purpose of the project is to involve trainees in actual multi-faceted civil engineering projects and to develop their team working skills. Participation requires methodology and communication skills, technical knowledge and design know-how. The trainees are expected to adopt a project-based approach in which a solution to the question is designed according to the steps of the design cycle in a defined time frame. Approach and result of the group project is presented to the supervisors and problem owners. The project is completed with a final report.	
Study Goals	After completion of the project the trainee should be able to: * Integrate and apply theoretical knowledge in complex and initially open-ended civil projects * Evaluate options and systematically make choices based on qualitative and quantitative arguments, working with conflicting requirements * Plan and organize a project with milestones and specifications of activities and tasks * Professionally work in a team and deal with sponsors, mandators and other stakeholders	
Education Method	Collectively run project under supervision of the programme coordinators and scientific supervisors.	
Assessment	The outcome is assessed by involved supervisors and the trainees on the basis of a defined set of criteria, including engineering and scientific quality, working attitude, obtained results, reporting and presentation.	
Remarks	Participants of the EngD CEE only.	
Expected prior Knowledge	MSc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	None	
Judgement	See Assessment	
Permitted Materials during Exam	None	
Collegerama	No	

CEE5006	Individual Design Project	60
Responsible Instructor	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	See timetable in Brightspace	
Education Period	1 2 3 4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The individual design project is the main part of the EngD programme. It is carried out in relation to the strategic innovation needs of the involved external party from the civil sector and supported by compulsory and elective courses. First the trainee needs to write a project letter including objective, approach and result of the project and has to prepare a presentation to the selection committee. After approval for admission to the programme by the selection committee, the trainee will start with courses and work as well on the individual project. After the first year, the results of the courses and the individual project are evaluated in a Go No Go meeting by the involved professors and supervisors. After a positive outcome, participation in the programme will be continued. At the end of the second year of the programme, a green light meeting will be held to evaluate the draft thesis and decide on approval for planning of the thesis defence.	
Study Goals	Through the execution of the individual project, the trainee will develop broad knowledge and skills for designing and developing innovations. In addition to the related scientific and engineering perspectives also the business, management and finance aspects of innovation are considered.	
Education Method	A substantial part of the knowledge will be developed in the compulsory and elective courses. In addition the involved supervisors from both TU Delft and the involved external party provide guidance during the execution of the project.	
Assessment	The individual project and the related thesis is evaluated by the involved EngD committee on the basis of a defined set of criteria.	
Remarks	Participants of the EngD CEE only.	
Expected prior Knowledge	MSc level, preferably in exact sciences	
Academic Skills		
Literature & Study Materials	None	
Judgement	See Assessment	
Permitted Materials during Exam	None	
Collegerama	No	

CEE5007	Techno-Economic Evaluation for Designers	6
Responsible Instructor	Dr. J.M. Vleugel	
Course Coordinator	Ing. C.H. Groenewegen	
Contact Hours / Week x/x/x/x	Depending on the content between 3-6 hours per week (see timetable in Brightspace)	
Education Period	3 4	
Start Education	4	
Exam Period	Exam by appointment	
Course Language	English	
Course Contents	The course focuses on the development of knowledge and skills to prepare business cases to support decision-making about the financing of development projects or innovation projects. The course consists of three modules in which knowledge about 1) Financial management 2) Technology strategy and 3) cost-benefit analysis is developed. The fourth and final module consists of writing, presenting and defending a business case about the individual EngD project of the trainee or a case from the industry.	
Study Goals	In Modules 1-3 the trainee learns to - Interpret and evaluate financial statements and financial performance of companies and or businesses. - Understand and analyse the industry dynamics of technological innovation and companies technology strategy. - Gain understanding of the fundamentals of cost-benefit analysis (CBA) and know the strength and limitations in project or innovation evaluation. - Apply relevant techniques and methods for decision making on investment in development and renewal projects. In Module 4 the trainee learns to 1) Consider several alternative solutions for a specific technical challenge; 2) Carry out an integral assessment of the individual design or research project including technical, financial, benefit, environmental, safety aspects and other relevant facets like user friendliness, health quality and sustainability; 3) Define development projects regarding technical feasibility, economic feasibility, regulation and reliability. Such a project should be managed within the (financial) budgets for people, resources and systems. Decisions and design choices should be based on solid reasoning; 4) Present the value of a project or innovation for the organisation and outline the required financial funding and resources for the implementation.	
Education Method	Class lectures and individual contact about the execution of the assignments and business case.	
Assessment	Execution of short assignments in the modules on technology strategy, financial management and cost benefit analysis. The assignments can be used as building blocks in the final assignment. Evaluation of an individual assignment is based on the written business case with a maximum of 12 pages and an oral presentation of 15 minutes followed by a short defence. The written part of the assignment should include: - A context analysis describing the business and industry essentials - Description of the business problem or opportunity - Discussion of the innovative aspects of the product under development (Technical) requirements, constraints and kpi (as far as relevant in the product development stage) - Description of a business as usual scenario and its consequences - An investment analysis (capex) necessary investment, funding sources, payback time - An operational analysis (opex) including maintenance and end-of-life (if relevant) - A comprehensive risk analysis (technical, financial, economic (demand), environmental, organisational aspects) - Required competences in the organisation to deliver the project outcomes - Impact on current operations (disturbances during implementation) - Implementation plan with a timescale - Remaining technical issues after implementation An assessment table is published on Brightspace	
Remarks	In principle this course is for participants of the TU Delft EngD programmes only. If you are a PhD student and would like to follow this course please ask the responsible instructor if you can join this course.	
Expected prior Knowledge	MSc level, preferably in exact sciences.	
Academic Skills		
Literature & Study Materials	Course materials as published on Brightspace and your own lecture notes.	
Judgement	See assessment.	
Permitted Materials during Exam	Your business case report.	
Collegerama	No	

Year	2024/2025
Organization	Civil Engineering and Geosciences
Education	EngD Civil and Environmental Engineering

PDEng CEE Elective courses

Dr.ir. M. van den Boomen

Department	Hydraulic Structures and Flood Risk
Room	23.HG 6.71
Department	Integral Design & Management
Room	23.HG 6.71
Department	Integral Design & Management
Room	23.HG 6.71
Department	Integral Design & Management
Room	23.HG 6.71

S. Cornelissen

Prof.dr.ir. P.H.A.J.M. van Gelder

Unit	Techniek, Bestuur & Management
Department	Safety and Security Science
Telephone	+31 15 27 86544
Room	31.c1.170

Ing. C.H. Groenewegen

Department	Railway Engineering
Telephone	+31 15 27 85653
Room	23.S2 1.33

Ir. W.G. van der Hoog

Department	Onderwijs Inhuur
Department	Onderwijs Inhuur
Department	Onderwijs en Studentenzaken

Dr. J.M. Vleugel

Unit	Civiele Techniek & Geowetensch
Department	Transport, Mobility and Logistics
Telephone	+31 15 27 86487
Room	23.HG 4.18

Dr. M. Yang

Department	Safety and Security Science
Telephone	+31 15 27 82891
Room	31.C1.130